

- Change the trace from a player shaped trace to a rectangle which curves to follow the player. Rather than having noticeable sections, make it one fluid shape. It should emanate from the middle of the player's model, and follow all the exact same rules as the old trace, but look cleaner. Think a Tron path but from the model's back, curving through the air.
- Perfectly timed parries should kill the enemy dashing. E.g. if an enemy's dash would kill the carrier, but the carrier stops it with a parry, that enemy dies instead. Retain all other rules regarding dashing a trace and parrying.
- For game mode b:
 - create a small invisible radius around players that acts as a "catch zone," so that when the core enters that area, it curves towards the player like a magnet. This is intended to make catching feel fluid and clean.
 - When a team has possession of the core, throws it, and it hits the ground, it should automatically turnover to the closest player on the enemy team.
 - Raise the goals 1.5x player height from the ground, place them into the back walls, and make them circular
- This version is WAY laggy, with 1/3 the fps and its unplayable. The host address isn't showing and there's no join option.
- The trailhead gap is still not fixed. When the player is carrying the core, there is a big gap between the player model and the most recent part of the trace.

Fix a build failure caused by two C4458 "declaration hides class member" errors in the Trace project (UE 5.8).

Error 1: Source/Trace/UI/TraceHUD.cpp, line 1602

`const TraceNet::ERole Role = TraceNet::DescribeConnection(GetWorld(), Endpoint, Detail);`
 This local `Role` shadows `AActor::Role` (inherited via `AHUD`). Rename the local variable to something like `ConnectionRole` and update all usages within its scope accordingly.

Error 2: Source/Trace/Core/TracePlayerController.cpp, line 564

`for (const APlayerState* Player : GameStateBase->PlayerArray)`
 This local `Player` shadows `APlayerController::Player`. Rename the loop variable to something like `PlayerStateEntry` and update all usages within the loop body accordingly.

After making both fixes:

1. Grep the rest of Source/Trace for any other local variables or loop variables named `Role`, `Player`, `Owner`, `Pawn`, or `Controller` inside Actor-derived classes (`AActor`, `APawn`, `APlayerController`, `AHUD`, etc.) that

could trigger the same C4458 warning-as-error, and rename any you find using the same pattern (descriptive suffix, no functional changes).

2. Run build.bat and confirm the build succeeds with no C4458 errors.
3. Report back a summary of every file and line changed.

Keep changes purely cosmetic (renames only) — no logic changes.